

Bio-inspired designs for robotic exoskeletons for Upper Body - a Review

Madhurya Mozumder

MSc. Robotics

University of the West of England

Madhurya2.Mozumder@live.uwe.ac.uk , ORCID: 0000-0001-7367-5736

Abstract—Robotic exoskeletons for the upper body hold immense potential in rehabilitation, assistive technologies, and human augmentation. Bio-inspired design principles offer a promising approach to overcome limitations in current exoskeleton technologies. This paper reviews the state-of-the-art in bio-inspired upper-body exoskeletons, focusing on advancements made since the year 2000. We explore the methodology used for literature selection and classification. The results section presents an organized review of findings, covering key areas such as mimicking skeletal structures, muscle actuation, and control strategies. The discussion section delves into the fundamental opportunities and challenges associated with biomimetic exoskeleton design. Finally, we draw conclusions and propose recommendations for future research directions.

Index Terms—Bio-inspired, exoskeleton, upper limb, rehabilitation, assistive technologies.

I. INTRODUCTION

Bio-inspired robotic exoskeletons for the upper body have brought about a major change in assistance and rehabilitation. A human arm has 7 Degrees-of-Freedom. The exoskeletons are designed to mimic the human musculoskeletal system, drawing inspiration from bones, muscles, and tendons [1]. Lightweight and often cable-driven, they use artificial muscles or motors to provide power [2]. This bio-mimetic design allows for a more natural movement experience, complementing human motion for tasks requiring strength or aiding in recovery from injuries [3]. By drawing inspiration from biology, researchers aim to develop exoskeletons that are:

- **Lightweight and adaptable:** Mimicking the lightweight yet robust structure of bones and the adaptability of muscles can lead to exoskeletons that conform better to human anatomy and movement patterns.
- **More natural and intuitive:** By replicating muscle actuation and control systems, bio-inspired exoskeletons can offer a more natural user experience and improve user comfort.
- **Energy-efficient:** Learning from the efficient energy expenditure of the human body can lead to exoskeletons with improved battery life and reduced power consumption.

For rehabilitation, bio-inspired exoskeletons are proving to be valuable tools for patients recovering from strokes, spinal cord injuries, or other neurological conditions. By providing controlled movement and amplifying weakened muscles, these

exoskeletons can significantly improve a patient's ability to regain motor function and independence [4]. Studies have shown that exoskeleton-assisted therapy can lead to faster and more effective recovery compared to traditional methods [5].

Beyond rehabilitation, bio-inspired exoskeletons are making a significant impact in industrial settings. By offering support during physically demanding tasks, such as lifting heavy objects or performing repetitive motions, exoskeletons can help reduce worker fatigue and prevent musculoskeletal disorders [6]. This translates to a safer and more productive work environment for employees.

The applications extend even further. Athletes are increasingly utilizing bio-inspired exoskeletons for performance enhancement. These exoskeletons can provide additional strength and endurance during training, allowing athletes to push their limits and achieve optimal results [7]. Imagine a baseball player swinging a bat with enhanced power or a weightlifter performing heavier lifts with improved stability.

The potential of bio-inspired upper limb exoskeletons continues to grow. Researchers are constantly developing new and improved designs with expanded functionalities. This technology holds immense promise for improving not only patient recovery and worker safety but also pushing the boundaries of human performance across various fields.

Bio-inspired robotic exoskeletons for the upper body offer a compelling blend of assistance and innovation, but they are not without limitations. On the positive side, these exoskeletons provide superior strength and endurance, amplifying the user's capabilities for tasks requiring heavy lifting or strenuous exertion [8]. In rehabilitation settings, they offer targeted therapeutic intervention, promoting motor function recovery and potentially accelerating the healing process [9]. Additionally, bio-inspired designs can improve natural movement patterns, making the exoskeleton feel more intuitive and comfortable for the wearer [10].

However, challenges also exist. The complexity of these machines can lead to high costs, limiting accessibility for some users [11]. Additionally, the bulk and weight of exoskeletons can restrict user movement and potentially cause discomfort during extended wear [12]. Finally, the technology is still under development, and the long-term effects of using exoskeletons for extended periods require further investigation [13].

Despite these drawbacks, bio-inspired upper limb exoskeletons represent a significant advancement in assistive technology. As research and development continue, we can expect to see these exoskeletons become more refined, affordable, and user-friendly, ultimately offering even greater benefits across various applications.

II. METHODOLOGY

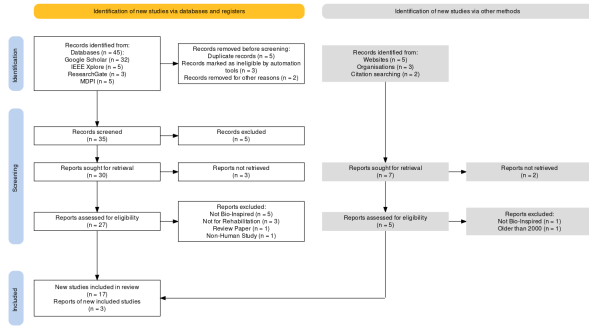


Fig. 1. PRISMA [14] flow diagram illustrating the number of reviewed articles through the different phases

A. Search Terms

Relevant search terms were used to match the research with the objectives to get relevant papers. The keywords that were used to get these papers can be listed as:

- **Targeted Body Segment:** The search focused on the upper body using terms like "arm," "shoulder," "elbow," and "wrist." Excluding terms related to the lower body ensured the results centered on upper-limb exoskeletons.
- **Actuation Type:** Keywords like "powered," "active," and "motorized" identified literature discussing exoskeletons with external power sources for movement assistance.
- **Assistance Focus:** Terms such as "aid," "assist," "augment," and "enhance" captured research aimed at improving human capabilities through exoskeleton technology.
- **Exoskeleton Terminology:** A range of terms encompassing "exoskeleton," "wearable robot," and "robotic suit" ensured the inclusion of relevant studies exploring various exoskeleton designs.
- **Control Strategy:** Keywords like "control" and "validation" identified research that delved into the control algorithms and validation methods used in bio-inspired upper-body exoskeletons.
- **Application Exclusion:** The term "post stroke" was excluded to narrow the focus on bio-inspired designs applicable to a broader range of upper-body exoskeleton applications beyond stroke rehabilitation.

B. Database and search engine selection

Google Scholar was used as a primary tool to look up the papers that match the keywords. It provided a vast collection of scholarly literature spanning across multiple theses, articles, books, and abstracts from various academic institutions like universities, online libraries, repositories and academic

publishing forums. Google Scholar however does not provide in-depth insight into the search and since it refers to multiple databases across the Internet, there are duplicates that appear in the search. This needs to be eliminated and secondary search engines like IEEE Xplore which are far less susceptible to duplicate entries have been used. Besides this, online forums like ResearchGate, PubMed, Web of Science and MDPI were referred to. They are specialized forums concentrated on relevant research niches, hence the search was more streamlined. However, since these forums are more disciplinarily narrow as compared to Google Scholar, the yield of papers was low.

C. Inclusion and Exclusion Criteria

The paper majorly focuses on a review of Bio-inspired designs for upper-body exoskeletons. The main criteria for including papers in the study was if these papers proposed some novel solution for upper body exoskeleton that was bio-inspired. The papers reviewed were grouped loosely into some categories for easier .The following categories were established:

- **Control Strategy:** Papers emphasizing control algorithms for bio-inspired exoskeletons were grouped under this category. These studies typically explored specific control strategies and their implementation in different exoskeleton designs.
- **System Overview:** Papers introducing the mechanical design, functionalities, and sensor and control systems of bio-inspired upper-body exoskeletons fell into this category.
- **Validation with Users:** This category included studies presenting and analyzing experiments conducted with human subjects wearing bio-inspired exoskeletons. Examining these papers provided insights into the effectiveness of different design principles on user performance and experience.
- **Other Validation Methods:** Papers addressing the mechanical design or sensor apparatus of bio-inspired exoskeletons, along with their characterizations, but without human-in-the-loop experiments, were classified here. These papers were considered supplementary unless they provided the only available experimental data for a specific exoskeleton.
- **Review Papers:** Existing review works on bio-inspired upper-body exoskeletons were included in this category. This comprised the screening process that resulted in 32 papers. These papers were brought to the second phase where a few papers were further excluded on the classification criteria
- **Not Bio-Inspired:** Papers with exoskeleton designs that were not inspired by biological morphologies,5, were excluded.
- **Not for Rehabilitation:** Papers that do not deal with exoskeletons that focus on rehabilitation,1, was excluded.
- **Review Paper:** Papers that were essentially a review of existing literature , 1, were excluded.
- **Non-Human Study:** The study involved assistive robotic upper body exoskeletons not involving humans.

Finally with all the exclusions, out of a total 32 retrieved papers, 20 were considered for the final study. The process of selection of the papers has been illustrated in the PRISMA [14] diagram in Figure 1.

TABLE I
SUMMARY OF REVIEWED EXOSKELETON STUDIES

No.	Title	Year	Category	Application
1	An upper-body exoskeleton with novel shoulder mechanisms	2017	Assistive	Shoulder joint assistance
2	RUPERT: Exoskeleton for arm rehabilitation	2008	Assistive	Assistance in Stroke rehabilitation
3	Pneumatic Muscle Actuators for Upper Limb Exoskeleton	2020	Rehabilitation	Rehabilitation by shoulder flexion
4	RehabRoby: Upper Limb Exoskeleton Design	2011	Rehabilitation	Stroke patient rehabilitation
5	MAHI EXO-II: Upper Extremity System	2014	Rehabilitation	Long-duration therapy for upper limb
6	Force Feedback Exoskeleton for Virtual Environments	2005	Augmentation	Enhancing mobility
7	Multi-modal Interface Controlled Exoskeleton	2021	Assistive	Muscular dystrophy assistance
8	RUPERT: Clinical and Home Therapy Exoskeleton	2016	Assistive	Repetitive task assistance
9	CRUX: Lightweight Robotic Exosuit	2017	Augmentation	Multi-joint strength augmentation
10	Soft Exoskeleton for Stroke Recovery	2018	Assistive	Stroke-induced complication reduction
11	CAREX: Cable Driven Arm Rehabilitation	2012	Rehabilitation	Neural rehabilitation support
12	ARMin III: Shoulder Actuation Therapy	2009	Rehabilitation	Stroke rehabilitation
13	Exoskeleton Co-Simulation for Rehabilitation	2016	Rehabilitation	Post-stroke motor dysfunction
14	Whole Arm Wearable Robotic Exoskeletons	2014	Assistive	Support for daily activities
15	Multi-joint Diagnosis and Therapy Robot	2013	Rehabilitation	Comprehensive stroke rehabilitation
16	Space Robots Control with Exoskeleton	2006	Augmentation	Control interface for space robotics
17	Exo4Work: Occupational Task Support	2022	Augmentation	Reducing musculoskeletal load
18	Wearable Exoskeleton Prototype Testing	2021	Rehabilitation	Neuromotor rehabilitation
19	Exoskeleton for Daily Living Assistance	2014	Assistive	Daily living activity assistance
20	New Conceptual Design of Upper Limb Exoskeleton	2021	Rehabilitation	Assistance in lifting objects

III. RESULTS

A total of 20 studies were included in the review. A majority of these papers were from conference proceedings organized by IEEE or were published in academic platforms of eminence like Springer and MDPI. The journal articles of interest in this review were published between 2011 and 2023. A wide range of topics were discussed in these papers, ranging from applications of upper body exoskeletons in the medical field to assistive technologies for patients with limited locomotion capabilities and also in industrial sectors. Amongst the papers that were selected, 60% of the papers were from the search engines like Google Scholar and IEEE Xplore, 25% of the papers were from conference proceedings and publications in online academic forums like Springer, ResearchGate and MDPI and 15% from citations of review papers.

The papers were classified roughly into three categories based on their applications, namely Rehabilitation, Assistive and Augmentation. These categories help to carry out the study of the various robots according to the applications they were designed for.

- **Rehabilitation:** These exoskeletons are majorly used for helping patients who have suffered from physical disabilities like cardiovascular diseases leading to strokes. These patients face slight to severe weaknesses in the

upper limbs. Affected people lose their motor skills, including the ability to catch and lift objects. In many scenarios abnormal conditions like shaking and lack in coordination have been observed. Medical solutions to these include various physical therapies which help to improve strength. Robotic solutions have been proposed for these scenarios. The studies that do the same have been grouped under this category.

- **Assistive:** These exoskeletons can be considered as an intermediary between Rehabilitation robots and Augmentation robots. They can not only provide assistance to patients that have suffered injuries, that have affected their muscles of the upper limbs, but also healthy individuals, who in the course of daily work require extra support for their limbs like in factory assembly work or in scenarios where the exoskeleton can replace various sports equipment. These type of exoskeletons provide passive support.
- **Augmentation:** The exoskeletons that fall under this category are developed to improve human activity in scenarios that require heavy working conditions. They essentially serve as a support for the workers. These exoskeletons focus on supporting the weight of the manipulated object.

In the papers that were considered for the review, 50% were under Rehabilitation, 30% were under Assistive and 20% were under Augmentation.

IV. DISCUSSION

The studies considered have been classified as mentioned above into Rehabilitation, Assistive and Augmentation.

A. Rehabilitation

Rehabilitation robots are designed to provide patients with various ailments therapy that will help them regain the functionality of the affected parts of their body. These robots aim to provide active-assisted and resistive-assistive therapy as seen in the work of Fatih Ozkul and Duygun Erol Barkana, (2011) where an upper body exoskeleton named "RhabRoby" [15] helps in the upper body rehabilitation for stroke patients by implementing the aforementioned therapy. Huang, Tu and He, (2016) proposes a system having 5 Degrees of Freedom using compliant pneumatic muscles [16] for regaining arm movement capabilities which has led to another work using the same concept of pneumatic actuators by Chen et al., (2020), who brings forward a system for shoulder flexion, abduction, adduction and rotation, having 4 Degrees of Freedom [17]. This system uses air pressure for the control of the pneumatic muscles. A pioneer in the stroke rehabilitation sector is a 4DoF robot named "ARMin III" [18] by Nef, Guidali and Riener (2009) which has gone through three iterations on its design. This system is inspired from the design of the ball and socket joint of the shoulder. Strokes can bring about motor dysfunction and this has been addressed by an exoskeleton [19] designed by Pineda-Rico et al. (2016), which boasts 8 Degrees of Freedom. This system has been simulated on OpenSim

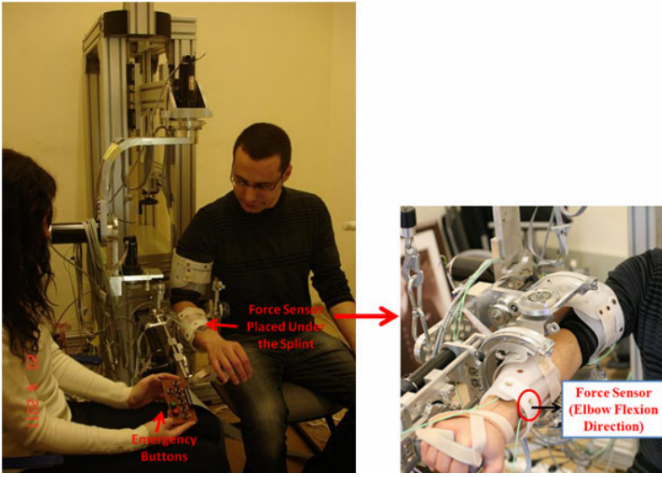


Fig. 2. RehabRoby with subject

[20], a simulation environment and holds promise to mature into a complex and extensive system for rehabilitation. Cable-driven systems mimic the natural functioning of the muscles and the tendons of the human arm. The first cable-driven system for the rehabilitation of neural impaired subjects named "CAREX" [21] was developed by Mao and Agrawal (2012), which boasted 5 Degrees of Freedom. This implementation was further used in the work by Ren et al. (2013) who designed the "IntelliArm" [22], that possessed 6 Degrees of Freedom, and has been tested effectively for stroke rehabilitation. It employs passive stretching and active movement to achieve its intended task. Progress in the modular design of the sensors

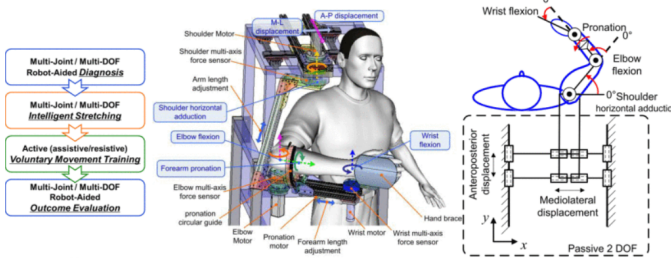


Fig. 3. Three-dimensional drawing of developed upper limb neurorehabilitation exoskeleton robot, the IntelliArm

brought about the incorporation of EEG and ECG systems into the exoskeletons as seen in the work of Vélez-Guerrero, Callejas-Cuervo and Mazzoleni (2021) who proposes a design having 1 Degree of Freedom for neuromotor rehabilitation for injuries [23]. The need of replicating the natural movements of the arm was satisfied by the work of Zakaryan, Harutyunyan and Sargsyan (2021) that allowed the users to lift heavy objects by exerting minimal effort [24]. However, in all these designs, a key component was not focussed on, that being the upper limb distal joint long duration therapy. This was the main focus of the design of French, Rose and O'Malley (2014) who proposed "MAHI Exo II" [25], a 5 Degree of Freedom robotic exoskeleton that helped in the rehabilitation of spinal cord

injury and stroke-afflicted patients. The designs proposed are novel in their own merit, each catering to the essential aspects of rehabilitation strategy in patients affected by dystrophy or injury due to stroke related complications. Rehabilitation robots essentially help in providing therapy to the afflicted patients by employing various strategies that assist in the physical therapies and exercises in a repetitive fashion.

B. Assistive

Assistive robots are designed to assist in various bodily functions in case of patients who have suffered from various injuries affecting their muscles and implicitly the biomechanics of their upper limb. Bai, Christensen and Islam, (2017) [1] designed a novel system with dual revolute joints, having 3 Degrees of Freedom that assist in the movement of the shoulder joints. A human arm has 7 Degrees of freedom and an assistive robot aims to supplement that fully or partially in order to help the patient achieve their daily living activities. This was the prime focus in the designs of Li et al. (2018) who designed a system with 7 degrees of freedom, possessing actuators that aligned with the natural muscle positions of the human arm [26], and Rahman et al. (2014) who designed the robot titled "ETS-MARSE" [27]. One of the early works in

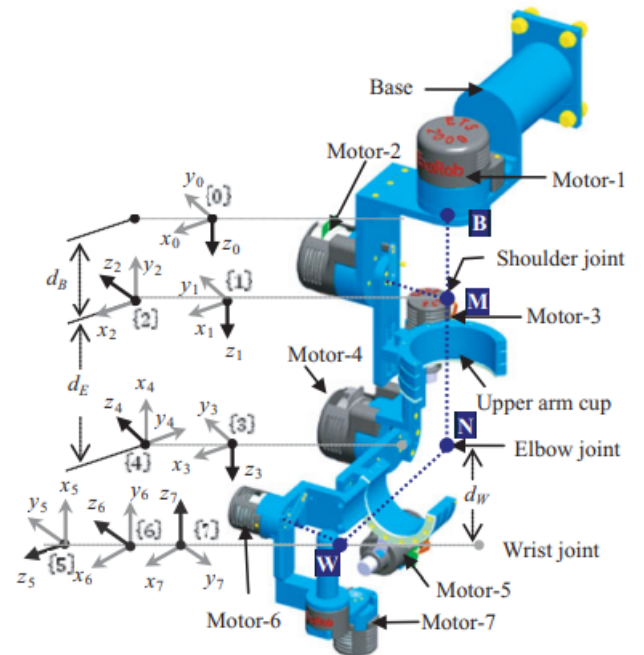


Fig. 4. Link frame attachments of the ETS-MARSE

this field was done by Balasubramanian et al. (2008) who designed the "RUPERT" [28] exoskeleton that has undergone four iterations on its design. One of the notable features of this system is its wearability and modularity, that is aimed at user comfort. The incorporation of EEG and EMG sensors to assistive exoskeletons was done by Win Tun Latt et al. (2014) in the robot named "SEXpo" [29] which leverages a

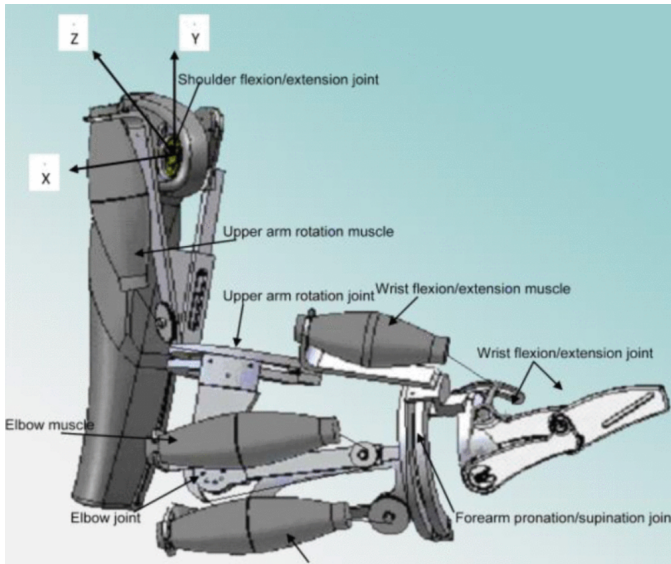


Fig. 5. Assembly of RUPERT

Microsoft Kinetic camera that is mounted on a wheelchair to aid in activities of daily living. Muscular dystrophy issues

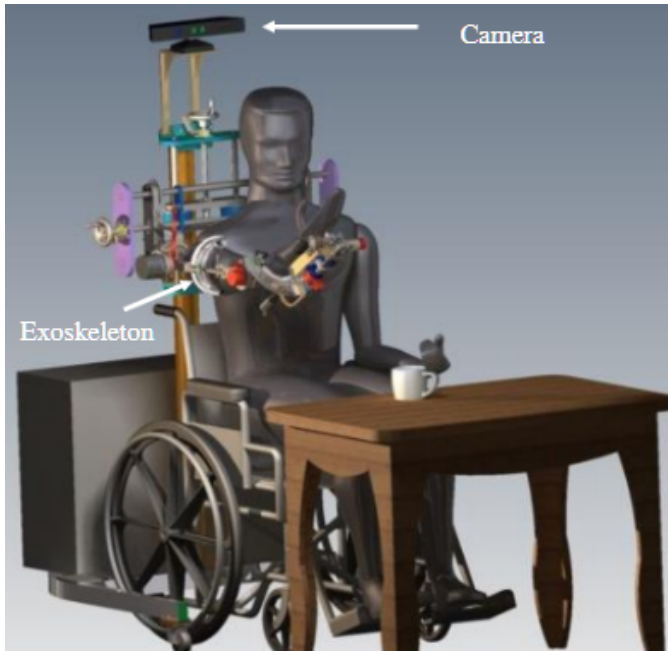


Fig. 6. An overall computer-aided design (CAD) of SPexo in an isometric view.

have been addressed in the design of the "BRIDGE" [30] exoskeleton by Gandolla et al. (2021). Assistive systems aim to provide support to the daily activities of the patients and with the increase in the complexity of the systems, the affected individuals receive a higher degree of support in achieving the daily activities.

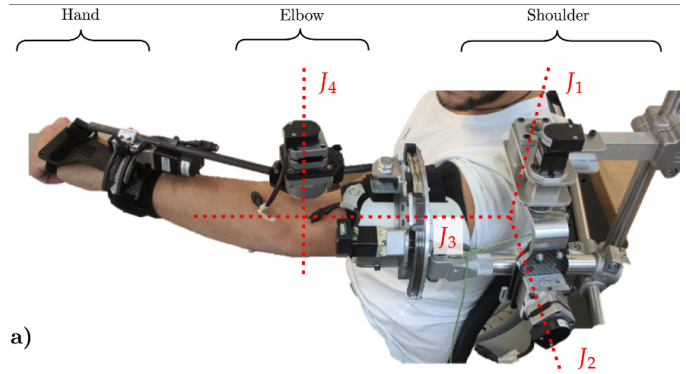


Fig. 7. The upper limb Bridge exoskeletal system.

C. Augmentation

Augmentation robots are not necessarily aimed at afflicted individuals, healthy individuals can also make use of these exoskeletons to achieve tasks that would otherwise require a lot of human effort to achieve. "L-Exos" [31], a tendon driven wearable haptic interface is one such system, designed by Frisoli et al. (2005) that possesses a novel Circular guide for enhanced mobility. The European Space Agency also has developed an augmentation system titled "EXARM" [32] that boasts 90 natural arm postures, facilitated by the use of 16 angular sensors. Inspired by the tensority models of the human arm, Lessard et al. (2017) developed a system named "CRUX Exosuit" [33] that provides strength amplification in multiple degrees of freedom. The human shoulder is subjected to a lot of stress whilst lifting heavy objects, and to ease that. Van der Have et al. (2022) developed "Exo4Work" [34] that enables reductions in musculoskeletal loading. Augmentation exoskeletons, however, not only have an implementation in enhancing the capabilities of the human body, but do play a pivotal role as seen in the application of "CRUX" and "L-Exos" in rehabilitation and Assisted living respectively.

V. CONCLUSION

Upper-limb exoskeletons hold significant promise for assisting people with movement and rehabilitation. This review article has explored the latest advancements in this field. The authors also briefly discussed hand exoskeletons, highlighting their potential integration with upper-limb devices. Readers interested in a deeper dive into hand exoskeletons can refer to the resources mentioned earlier.

The article discusses several novel approaches in upper-limb exoskeleton development. Currently, the mechanical design of upper-limb exoskeletons varies greatly due to the wide range of applications they target. There is no single established standard for designing and evaluating these robotic systems. Further research and development are needed to create novel mechanisms that better reflect the complexities of human biomechanics, particularly in the shoulder and wrist joints. Additionally, developing standardized evaluation criteria for control methods would allow for better comparisons between different studies.

REFERENCES

- [1] S. Bai, S. Christensen, and M. R. U. Islam, "An upper-body exoskeleton with a novel shoulder mechanism for assistive applications," in *2017 IEEE international conference on advanced intelligent mechatronics (AIM)*. IEEE, 2017, pp. 1041–1046.
- [2] H. I. Muri, L. Hoang, and D. R. Hjelm, "Mapping nanoparticles in hydrogels: a comparison of preparation methods for electron microscopy," *Applied Sciences*, vol. 8, no. 12, p. 2446, 2018.
- [3] E. Bardi, M. Gandolla, F. Braghin, F. Resta, A. L. Pedrocchi, and E. Ambrosini, "Upper limb soft robotic wearable devices: a systematic review," *Journal of NeuroEngineering and Rehabilitation*, vol. 19, no. 1, p. 87, 2022.
- [4] L. M. Vaca Benitez, M. Tabie, N. Will, S. Schmidt, M. Jordan, E. A. Kirchner *et al.*, "Exoskeleton technology in rehabilitation: Towards an emg-based orthosis system for upper limb neuromotor rehabilitation," *Journal of Robotics*, vol. 2013, 2013.
- [5] A. Frisoli, M. Barsotti, E. Sotgiu, G. Lamola, C. Procopio, and C. Chisari, "A randomized clinical control study on the efficacy of three-dimensional upper limb robotic exoskeleton training in chronic stroke," *Journal of neuroengineering and rehabilitation*, vol. 19, no. 1, p. 14, 2022.
- [6] D. Sui, J. Fan, H. Jin, X. Cai, J. Zhao, and Y. Zhu, "Design of a wearable upper-limb exoskeleton for activities assistance of daily living," in *2017 IEEE international conference on advanced intelligent mechatronics (AIM)*. IEEE, 2017, pp. 845–850.
- [7] S. Parasuraman, A. W. Oyong, and V. Ganapathy, "Development of robot assisted stroke rehabilitation system of human upper limb," in *2009 IEEE International Conference on Automation Science and Engineering*. IEEE, 2009, pp. 256–261.
- [8] A. Zeiaee, R. S. Zarrin, A. Eib, R. Langari, and R. Tafreshi, "Cleverarm: A lightweight and compact exoskeleton for upper-limb rehabilitation," *IEEE Robotics and Automation Letters*, vol. 7, no. 2, pp. 1880–1887, 2021.
- [9] D. Su, Z. Hu, J. Wu, P. Shang, and Z. Luo, "Review of adaptive control for stroke lower limb exoskeleton rehabilitation robot based on motion intention recognition," *Frontiers in Neurorobotics*, vol. 17, p. 1186175, 2023.
- [10] A. Gupta, A. Singh, V. Verma, A. K. Mondal, and M. K. Gupta, "Developments and clinical evaluations of robotic exoskeleton technology for human upper-limb rehabilitation," *Advanced Robotics*, vol. 34, no. 15, pp. 1023–1040, 2020.
- [11] N. Rehmat, J. Zuo, W. Meng, Q. Liu, S. Q. Xie, and H. Liang, "Upper limb rehabilitation using robotic exoskeleton systems: A systematic review," *International Journal of Intelligent Robotics and Applications*, vol. 2, pp. 283–295, 2018.
- [12] I. Aprile, A. Cruciani, M. Germanotta, V. Gower, C. Pecchioli, D. Cattaneo, F. Vannetti, L. Padua, and F. Gramatica, "Upper limb robotics in rehabilitation: an approach to select the devices, based on rehabilitation aims, and their evaluation in a feasibility study," *Applied Sciences*, vol. 9, no. 18, p. 3920, 2019.
- [13] M. Franceschini, S. Mazzoleni, M. Goffredo, S. Pournajaf, D. Galafate, S. Criscuolo, M. Agosti, and F. Posteraro, "Upper limb robot-assisted rehabilitation versus physical therapy on subacute stroke patients: A follow-up study," *Journal of bodywork and movement therapies*, vol. 24, no. 1, pp. 194–198, 2020.
- [14] N. R. Haddaway, M. J. Page, C. C. Pritchard, and L. A. McGuinness, "Prisma2020: An r package and shiny app for producing prisma 2020-compliant flow diagrams, with interactivity for optimised digital transparency and open synthesis," p. e1230, jun 2022. [Online]. Available: <https://doi.org/10.1002/cl2.1230>
- [15] F. Ozkul and D. E. Barkana, "Design and control of an upper limb exoskeleton robot rehabroby," in *Towards Autonomous Robotic Systems: 12th Annual Conference, TAROS 2011, Sheffield, UK, August 31–September 2, 2011. Proceedings 12*. Springer, 2011, pp. 125–136.
- [16] J. Huang, X. Tu, and J. He, "Design and evaluation of the rupert wearable upper extremity exoskeleton robot for clinical and in-home therapies," *IEEE transactions on systems, man, and cybernetics: systems*, vol. 46, no. 7, pp. 926–935, 2015.
- [17] C.-T. Chen, W.-Y. Lien, C.-T. Chen, and Y.-C. Wu, "Implementation of an upper-limb exoskeleton robot driven by pneumatic muscle actuators for rehabilitation," in *Actuators*, vol. 9, no. 4. MDPI, 2020, p. 106.
- [18] T. Nef, M. Guidali, and R. Riener, "Armin iii—arm therapy exoskeleton with an ergonomic shoulder actuation," *Applied Bionics and Biomechanics*, vol. 6, no. 2, pp. 127–142, 2009.
- [19] Z. Pineda-Rico, J. A. Sanchez de Lucio, F. J. Martinez Lopez, and P. Cruz, "Design of an exoskeleton for upper limb robot-assisted rehabilitation based on co-simulation," *Journal of Vibroengineering*, vol. 18, no. 5, pp. 3269–3278, 2016.
- [20] S. L. Delp, F. C. Anderson, A. S. Arnold, P. Loan, A. Habib, C. T. John, E. Guendelman, and D. G. Thelen, "Opensim: open-source software to create and analyze dynamic simulations of movement," *IEEE transactions on biomedical engineering*, vol. 54, no. 11, pp. 1940–1950, 2007.
- [21] Y. Mao and S. K. Agrawal, "Design of a cable-driven arm exoskeleton (carex) for neural rehabilitation," *IEEE transactions on robotics*, vol. 28, no. 4, pp. 922–931, 2012.
- [22] Y. Ren, S. H. Kang, H.-S. Park, Y.-N. Wu, and L.-Q. Zhang, "Developing a multi-joint upper limb exoskeleton robot for diagnosis, therapy, and outcome evaluation in neurorehabilitation," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 21, no. 3, pp. 490–499, 2012.
- [23] M. A. Vélez-Guerrero, M. Callejas-Cuervo, and S. Mazzoleni, "Design, development, and testing of an intelligent wearable robotic exoskeleton prototype for upper limb rehabilitation," *Sensors*, vol. 21, no. 16, p. 5411, 2021.
- [24] N. Zakaryan, M. Harutyunyan, and Y. Sargsyan, "Bio-inspired conceptual mechanical design and control of a new human upper limb exoskeleton," *Robotics*, vol. 10, no. 4, p. 123, 2021.
- [25] J. A. French, C. G. Rose, and M. K. O'malley, "System characterization of mahi exo-ii: A robotic exoskeleton for upper extremity rehabilitation," in *Dynamic Systems and Control Conference*, vol. 46209. American Society of Mechanical Engineers, 2014, p. V003T43A006.
- [26] N. Li, T. Yang, P. Yu, J. Chang, L. Zhao, X. Zhao, I. H. Elhajj, N. Xi, and L. Liu, "Bio-inspired upper limb soft exoskeleton to reduce stroke-induced complications," *Bioinspiration & biomimetics*, vol. 13, no. 6, p. 066001, 2018.
- [27] M. H. Rahman, M. J. Rahman, O. Cristobal, M. Saad, J.-P. Kenné, and P. S. Archambault, "Development of a whole arm wearable robotic exoskeleton for rehabilitation and to assist upper limb movements," *Robotica*, vol. 33, no. 1, pp. 19–39, 2015.
- [28] S. Balasubramanian, R. Wei, M. Perez, B. Shepard, E. Koeneman, J. Koeneman, and J. He, "Rupert: An exoskeleton robot for assisting rehabilitation of arm functions," in *2008 virtual rehabilitation*. IEEE, 2008, pp. 163–167.
- [29] W. T. Latt, T. P. Luu, C. Kuah, and A. W. Tech, "Towards an upper-limb exoskeleton system for assistance in activities of daily living (adls)," in *Proceedings of the International Convention on Rehabilitation Engineering & Assistive Technology*, 2014, pp. 1–4.
- [30] M. Gandolla, S. Dalla Gasperina, V. Longatelli, A. Manti, L. Aquilante, M. G. D'Angelo, E. Biffi, E. Diella, F. Molteni, M. Rossini *et al.*, "An assistive upper-limb exoskeleton controlled by multi-modal interfaces for severely impaired patients: development and experimental assessment," *Robotics and Autonomous Systems*, vol. 143, p. 103822, 2021.
- [31] A. Frisoli, F. Rocchi, S. Marcheschi, A. Dettori, F. Salsedo, and M. Bergamasco, "A new force-feedback arm exoskeleton for haptic interaction in virtual environments," in *first joint eurohaptics conference and symposium on haptic interfaces for virtual environment and teleoperator systems. World haptics conference*. IEEE, 2005, pp. 195–201.
- [32] A. Schiele, "8.3 case study: The ergonomic exarm exoskeleton," *Wearable robots: Biomechatronic exoskeletons*, p. 248, 2008.
- [33] S. Lessard, P. Pansodtee, A. Robbins, L. B. Baltaxe-Admony, J. M. Trombadore, M. Teodorescu, A. Agogino, and S. Kurniawan, "Crux: A compliant robotic upper-extremity exosuit for lightweight, portable, multi-joint muscular augmentation," in *2017 international conference on rehabilitation robotics (ICORR)*. IEEE, 2017, pp. 1633–1638.
- [34] A. van der Have, M. Rossini, C. Rodriguez-Guerrero, S. Van Rossum, and I. Jonkers, "The exo4work shoulder exoskeleton effectively reduces muscle and joint loading during simulated occupational tasks above shoulder height," *Applied Ergonomics*, vol. 103, p. 103800, 2022.